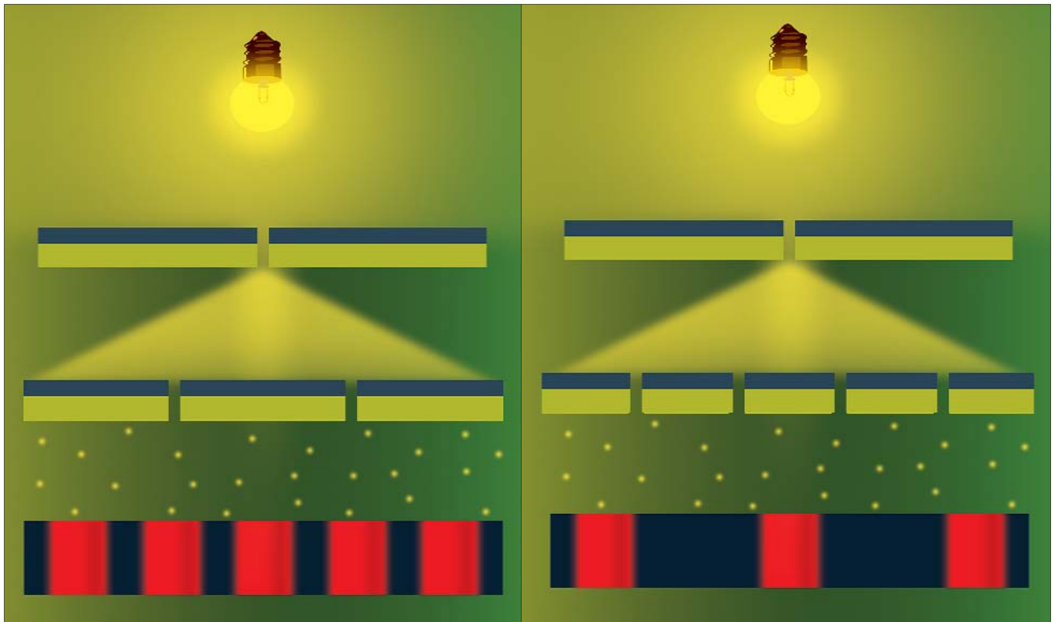
So What?

We're getting there. We're trying to explain how "a quantum computer" is gazillions of times faster than a silicon computer. The concept of the multiverse comes in here, according to David Deutsch, who laid the theoretical foundations of quantum computing.

An aside: we must mention that not everyone believes there are gazillions of universes out there. But we're subscribing to that view for two reasons—one, Deutsch insists upon it, and we must respect that insistence; and two, there really does seem no other way *Shor's Algorithm* (up soon) would work.



On the left, light shines through a smaill slit (so it's like a "point source" of light), then passes through an opaque screen with two slits. The pattern you get is as in the red and black space—red for bright, black for blank. On the right, just increase the number of slits to four, and the pattern changes! *What* is blocking the light?

Now, think of the problem of factorising large numbers. When you have two numbers such as 58359597 and 44845355, their product is 2617156845121935—easy enough to calculate. But when you have the latter number, how do you calculate what numbers it came from? In fact, factorising a 250-digit number could take longer than the known age of the universe on a regular computer! But Shor's algorithm—which is, well, an algorithm, devised by a man called, well, Peter Shor—actually does it in a reasonable amount of time. This, as it turns out, uses 10^{500} times the number of computational resources that seem to be present.

Deutsch asks, how can this have been achieved, when there are only 10^{80} atoms in the known universe?

The answer lies in "interfering universes." 10^{500} universes collaborate in the production of the answer, each of them doing a small part of the job. It's not like the problem is broken up into parts and given to each universe—there's no God out there to do that—it's just that this process is like the interference of light we talked about earlier. There, photons from other universes interfered with "our" photons, undetectable except for the fact that they interfered. Here, it's a mish-mash of answers from various universes that are interfering, leaving only one.

Now this might sound like rambling, and this writer might seem to be off his rocker, but that's common when dealing with quantum computing.

At this point you're probably screaming, "What *is* quantum computing?" Or "Just give me a description of a Quantel Inside processor!"

Sorry. We can't. That's just the way it is!

OK, we *can* shed some light on what's going on in the labs. For example, researchers have demonstrated cryptography working on quantum principles. Let's go through some real-world stuff.

Quantum Vagueness

Now we're getting into what a quantum computing environment looks like, what goes on, and how we get the answers.

The building blocks of a quantum computer are qubits instead of bits. A qubit can be not just one or zero, but can be both one *and* zero at the same time. So, if you have 10 bits, you can have only one of 2^{10} values; if you have 10 qubits, you can have all 2^{10} values at the same time. The qubit is based on the central ambiguity inherent in quantum mechanics: a property (such as spin) of a particle (such as an electron) is ambiguous until a process of disambiguation—such as an observation—causes the particle to "decide" upon what properties it has. Electron spin is too complex to go into here; just think of it as a property!

What does a qubit look like? Well, what does a bit look like? It's a voltage in a transistor, in most cases! And if that isn't hard to visualise, neither should a qubit be hard to visualise—qubits are represented by a quantum mechanical property of a particle such as an electron. They need to be set up such that they are in both states of spin at the same time. When one observes the state of the electron, the ambiguity vanishes, and this is called quantum decoherence.

An aside: we referred to a cat being both dead and alive until observed. That is one of the most popular paradoxes in quantum mechanics—what is the cat like until it is observed, and why should our observation make a difference? This is called the Schrödinger's Cat paradox, and if you've been reading keenly thus far, you should straightaway get yourselves a copy of *In Search Of Schrödinger's Cat*, by eminent physicist John Gribbin.

Now comes the meat of the matter: feeding in the input and getting the results! We need to present a series of qubits with a problem, and also a way to test the answer. This is done by setting up the quantum decoherence of the qubits such that